Dakota Access Pipeline: Financial Responsibility for a Potential Oil Spill at Lake Oahe

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Dakota Access Pipeline: Financial Responsibility for a Potential Oil Spill at Lake Oahe

The Dakota Access Pipeline (DAPL) carries crude oil beneath Lake Oahe, near the Standing Rock Sioux Tribe. The crossing raises an economic question: if an oil spill occurs, who would be responsible for paying for damage to the water and surrounding communities? I recommend that the U.S. Army Corps of Engineers first verify DAPL's existing financial protection before deciding whether additional insurance, a bond, or another form of financial security is necessary.

Background

The U.S. Army Corps of Engineers (2025) estimates that a worst-case oil spill at Lake Oahe could release 16,996 barrels. The Environmental Impact Statement (EIS) identifies 14 water intakes along 160 river miles. Although the EIS does not predict that the Standing Rock Sioux Tribe's drinking-water intake would be affected within the ten-day modeling period, it identifies potential impacts on agricultural water intakes (U.S. Army Corps of Engineers, 2025).

The EIS states that Dakota Access "maintains financial responsibility for the duration of the response actions," but this does not tell us how much insurance or bond coverage the company has (U.S. Army Corps of Engineers, 2025). The Corps may require a bond or other financial security, but I could not find a confirmed coverage amount (U.S. Army Corps of Engineers, 2025). I used the federal onshore-facility liability figure of $725,710,800 as a comparison, not as proof of DAPL's coverage or the legal limit applicable to this crossing (33 C.F.R. § 138.230(c)). Federal law also provides exceptions to liability limits, including gross negligence (33 U.S.C. § 2704).

Analysis

Negative externalities occur when an activity creates costs for people who are not directly involved in the decision. If an oil spill affects agricultural water intakes near Lake Oahe, farmers and surrounding communities could face financial losses even though they did not decide to operate the pipeline. If DAPL cannot fully pay for the damages, some costs could fall on these communities. Financial assurance could help internalize these costs, meaning the company prepares to cover potential harm from its operations. A liability limit could also create moral hazard if it reduces the company's incentive to prevent spills, although legal exceptions may apply.

Another concept is expected loss, calculated by multiplying the probability of an event by its estimated cost. For the 1.42-mile project-area bore, including the Lake Oahe crossing, the EIS gives a historical return period of 744 years for spills of any size. After adjusting for the pipeline's design, it estimates 37,207 years for spills of any size and 5,774,148 years for spills exceeding 10,000 barrels (U.S. Army Corps of Engineers, 2025). These estimates measure different events and should not be treated as the same probability. The EIS suggests some estimates may overstate actual spill frequency, while a commenter argues that the analysis underestimates the risk (U.S. Army Corps of Engineers, 2025).

To estimate potential damages, I compared two previous oil spills. The Kalamazoo River spill cost approximately $1.039 billion, or about $55,000 per barrel (“The Official Price,” 2013). The Yellowstone River spill had reported costs of approximately $135 million for about 1,000 barrels, or $135,000 per barrel (Associated Press, 2011). Applying these rates to the 16,996-barrel worst-case discharge gives an estimated damage range of $934.78 million to $2.29

billion. These are rough comparisons because neither spill occurred at Lake Oahe, and cleanup costs do not necessarily increase at the same rate for every additional barrel.

Using the design-adjusted return period for spills exceeding 10,000 barrels, my simplified annual expected loss ranges from approximately $162 per year using the $934.78 million damage estimate to $397 per year using the $2.29 billion estimate. These small averages reflect the extremely low estimated probability, not the actual cost if a major spill occurs. Figure 1 compares estimated damages with the $725.7 million federal liability figure. Based on the graph's assumptions, damages would exceed this amount at approximately 13,195 barrels using the Kalamazoo rate or 5,376 barrels using the Yellowstone rate.

Discussion

If the federal liability limit applies and damages exceed it, legal exceptions and other possible sources of compensation become important. If DAPL already has enough verified coverage, additional security may not be necessary. If its coverage is insufficient, the Corps should consider whether it can legally require more. For example, the $1.039 billion cost of the Kalamazoo spill exceeds the $725.7 million liability comparison by approximately $313.3 million. This does not mean DAPL would have an unpaid balance of that amount. The company could still be responsible for additional damages, and the Oil Spill Liability Trust Fund may cover certain eligible claims (33 U.S.C. § 2712).

One of the strongest arguments against additional financial security is that a major oil spill is extremely unlikely. My estimated annual expected loss of $162–$397 per year supports this argument, especially because insurance or bond coverage could increase operating costs. I cannot determine a reasonable insurance premium without more information. My concern is not

that a major spill is likely, but that the financial consequences could be substantial if one occurs. The pipeline's economic benefits also matter. The Three Affiliated Tribes estimated that shutting down the pipeline for one year could result in more than $160 million in losses (U.S. Army Corps of Engineers, 2025). Reviewing DAPL's financial coverage does not necessarily mean shutting down the pipeline.

Proposed Response

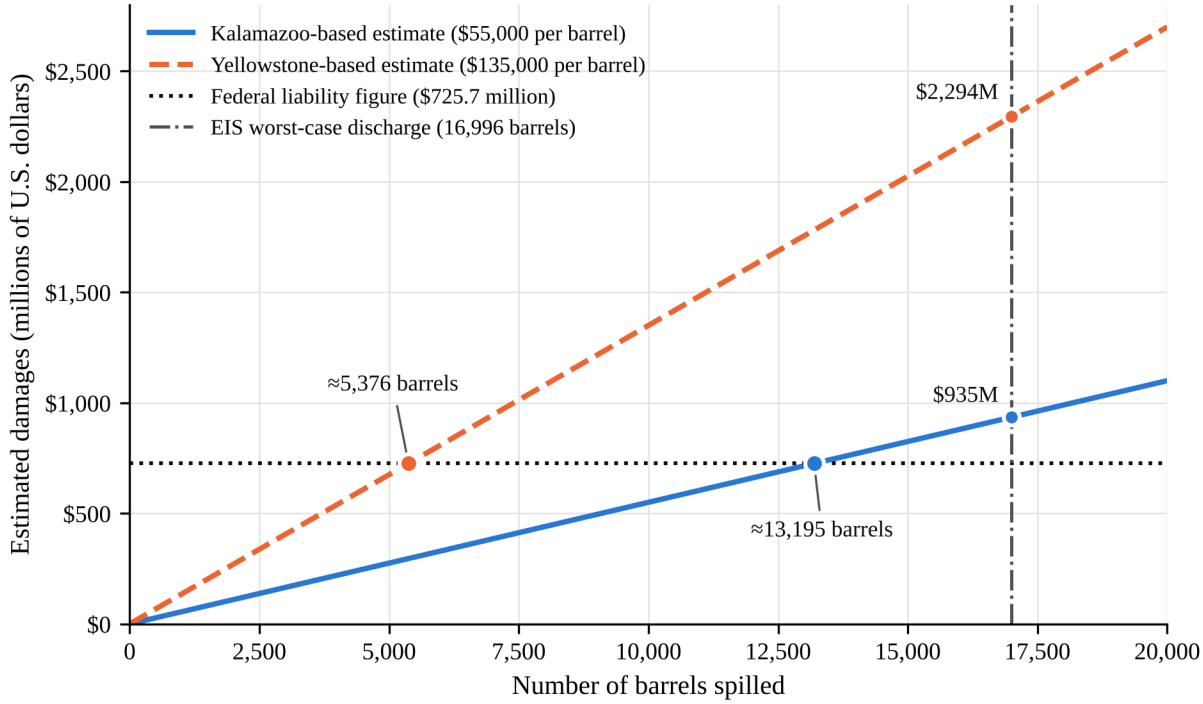
My recommendation is that the Army Corps first require independently verified information about DAPL's existing insurance, bonds, or other financial security. If the Corps has the legal authority and the company's coverage is insufficient, it should consider additional assurance based on a Lake Oahe-specific damage assessment. My estimated range of $934.78 million to $2.29 billion could be a starting point for that review, not a final requirement. By October 2027, I would reconsider my recommendation if independently verified information showed that DAPL already has enough financial protection to cover a major spill or if stronger evidence showed that my damage estimates were substantially too high.

Conclusion

Although a major oil spill at Lake Oahe may be extremely unlikely, its financial consequences could still be significant. Because DAPL's actual coverage has not been verified in the materials I reviewed, I think the Corps should confirm that coverage before deciding whether additional financial security is necessary.

Figure 1

Estimated Oil Spill Damages Compared With the Federal Liability Figure



Note. Author's calculations; straight lines assume a fixed cost per barrel. Worst-case discharge from the U.S. Army Corps of Engineers (2025); liability figure from 33 C.F.R. § 138.230(c); cost per barrel from The Official Price (2013) and Associated Press (2011). The liability figure is not DAPL's confirmed insurance or financial assurance.

References

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